

Trainings ▾

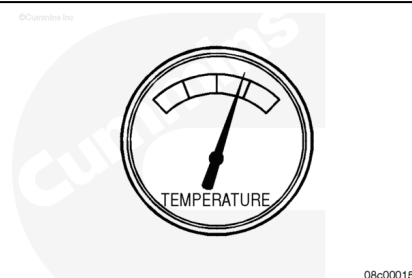
General Information

This engine is equipped with two thermostats. These thermostats open at different temperatures to improve cooling system efficiency. The low-temperature thermostat opens between 82°C [180°F] and 95°C [203°F]. The high-temperature thermostat opens between 88°C [190°F] and 97°C [207°F].

The functions of the cooling system are to:

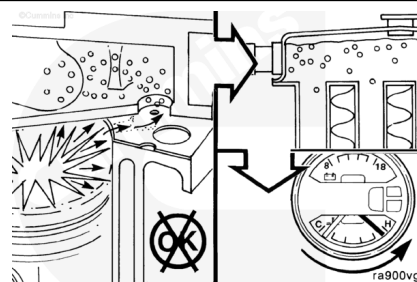
- Absorb heat from the engine components.
- Circulate the coolant in the engine and aftertreatment components.
- Control coolant temperature by the thermostat.

The coolant **must** be made up of the correct proportions of water, antifreeze, and Supplemental Coolant Additive (SCA) to perform these functions properly.



Air in the coolant can result in loss of coolant from the overflow when the aerated coolant is hot. The heated air expands, increasing the pressure in the system, causing the cap to open.

Similarly, coolant can be displaced through the overflow if the head gasket leaks compression gas into the coolant system.

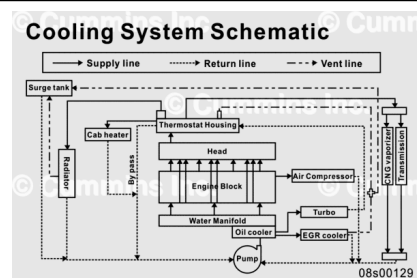


This illustration shows the coolant flow through the engine.

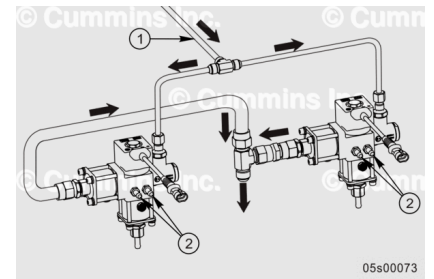
The arrangement of coolant thermostat housing and pipes are different due to the different applications that are with or without retarder.

Use the following procedure for more detail. Refer to Procedure 200-003 in Section F.

(/qs3/pubsys2/xml/en/procedures/1016/1016-200-003.html)



For Compressed Natural Gas (CNG) engine, there are two same fuel pressure regulators on this product, installed in parallel for fuel flow (1). Each one is equipped with two coolant ports (2).



The following publications, available through a Cummins® Authorized Repair Location, provide cooling system installation recommendations and specifications approved by Cummins Inc.:

- Data Sheets for specific engine models.
- Refer to Service Bulletin, Operation of Automotive Natural Gas Engines in Cold Climates, Bulletin 4332709 ([/qs3/pubsys2/xml/en/bulletin/4332709.html](http://qs3/pubsys2/xml/en/bulletin/4332709.html)).
- See Fluids for Cummins® Products Service Manual, Bulletin 5411406.

